

# Подстановка $x = 2 \cos \theta$ и многочлены Чебышёва

Домашнее задание

1 сентября 2026 года

**Теоретический минимум.**  $T_n(\cos \theta) = \cos n\theta$ ,  $U_{n-1}(\cos \theta) = \sin n\theta / \sin \theta$ , the recurrence  $T_{n+1} = 2xT_n - T_{n-1}$  and the hyperbolic branch  $T_n(\cosh t) = \cosh nt$ ; Chebyshev's extremal theorem; the inequalities of Markov and Bernstein; orthogonality with weight  $(1 - x^2)^{-1/2}$ ; the semicircle substitution  $x = 2 \cos \theta$  in three-term recurrences.

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1. Prove that  $T_m \circ T_n = T_{mn}$  and that these are, up to affine conjugacy, the only polynomial families of degree  $> 1$  closed under composition together with the powers  $x^n$  (теорема Ритта, достаточно сформулировать и проверить оба примера).
2. Prove the orthogonality relations  $\int_{-1}^1 \frac{T_m T_n}{\sqrt{1-x^2}} dx = 0$  for  $m \neq n$ , and expand  $x^n$  in the basis  $T_0, \dots, T_n$ .
3. Prove that the roots of  $U_{n-1}$  are  $\cos \frac{k\pi}{n}$  and deduce that the eigenvalues of the tridiagonal matrix with 0 on the diagonal and 1 on the neighbouring diagonals are  $2 \cos \frac{k\pi}{n+1}$ ; compare with the discrete Wirtinger inequality from the first sheet.
4. Prove that  $\min_p \max_{[a,b]} |p| = 2\left(\frac{b-a}{4}\right)^n$  over monic  $p$  of degree  $n$ , and find the polynomial attaining it.
5. (Proposed problems, №235) Prove that  $\cos \frac{\pi}{7}$  has minimal polynomial  $x^3 - \frac{1}{2}x^2 - \frac{1}{2}x + \frac{1}{8}$  over  $\mathbb{Q}$ , that  $\cos \frac{2\pi}{7}$  has minimal polynomial  $x^3 + \frac{1}{2}x^2 - \frac{1}{2}x - \frac{1}{8}$ , and that  $\cos \frac{3\pi}{7}$  has the same minimal polynomial as  $\cos \frac{\pi}{7}$ .
6. Prove that  $2 \cos \frac{2\pi}{n}$  is a rational number only for  $n \in \{1, 2, 3, 4, 6\}$ , using the recurrence  $T_{n+1} = 2xT_n - T_{n-1}$  and the rational root theorem.

7. Let  $p$  be a real polynomial of degree  $n$  with  $|p(x)| \leq 1$  on  $[-1, 1]$ . Prove that  $|p(x)| \leq |T_n(x)|$  for every  $|x| > 1$ , and that the leading coefficient of  $p$  is at most  $2^{n-1}$  in absolute value.
8. Prove that  $\sum_{k=0}^n T_k(x)t^k = \frac{1-tx}{1-2tx+t^2} + O(t^{n+1})$ , i.e. find the generating functions of  $T_n$  and  $U_n$ , and deduce the addition formula  $T_{m+n} + T_{m-n} = 2T_mT_n$ .
9. (Solve in radicals) Solve the equation  $x^3 - 3x = c$  for every real  $c$  by the substitution  $x = 2\cos\theta$  when  $|c| \leq 2$  and  $x = 2\cosh t$  otherwise; explain what the casus irreducibilis of Cardano's formula looks like in this language.
10. (IMC 2026) Prove that there exists a constant  $C > 0$  such that for every pair  $A, B$  of positive integers there is a real polynomial  $p(x)$  with

$$p(0)^2 > \sum_{i=1}^A p(-i)^2 + \sum_{i=1}^B p(i)^2 \quad \text{and} \quad \deg p < C\sqrt{AB}.$$